

User Guide Voiland Food Pantry Program

Version: Draft 4

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Date: 5 - 7 - 26

Introduction

This document provides instructions for operating the ID card scanner, the barcode scanning system and offline terminal program. It is intended for end users and clients to understand system functionality and basic operations. Included is a section for future testers and developers who want to improve the program or deploy it online.

System Overview

The system includes the following components:

- Software application for data entry and management
 - Card swipe/ID reader for user identification
 - Handheld barcode scanner for item input
 - Database for storing user and item records
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STUDENT USER GUIDE

First-Time Student (Registration)

Note: Registration is optional if logging in as a Guest

Click on the Student Option to access the Guest login if card is not working

Step 1: Arrive at the system

- You will be identified (likely by ID scan or input).
- If you are new, the Registration Popup appears.

Step 2: Complete Registration

1. Select your Class Year
 - Freshman / Sophomore / Junior / Senior
2. Enter your Major
3. Click “Complete Registration”

Your account is now created in the system.

Returning Student (Check-Out Process)

Step 1: Welcome Screen

- You’ll see:
 - “Welcome back”
 - Empty scanned items list

Step 2: Add Items

1. Click “+ Scan/Type Item”
 2. Enter or scan the product code (UPC)
 3. Click Add
- If valid → item name appears
 - If invalid → error message

Step 3: Confirm Item

1. Review item name
2. Click “Confirm Add”

Item is added to your list

Repeat until done.

Step 4: Submit Visit

1. Click “Submit Visit”
 2. System:
 - Saves items
 - Records visit with timestamp
 - Links visit to your student profile
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VOLUNTEER USER GUIDE

Register as Volunteer

Step 1:

- Go to Volunteer Register Page

Step 2:

Fill out:

- Full Name
- Email
- Password

Step 3:

Click Register

Account is created in the database

Login

1. Enter:

- Email
- Password

2. Click Login

Volunteer Dashboard

After login, you'll see:

Features:

- Welcome message
- Total hours worked
- Recent submissions

Submit Volunteer Hours

Step 1:

Enter:

- Hours worked (0–8)
- Date (limited to last 2 weeks)

Step 2:

Click Submit Hours

System stores:

- Hours
 - Date
 - Linked volunteer
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ADMIN USER GUIDE

Default Login

When system starts, a default admin is created:

- Username: admin
- Password: password123

If this rather dangerous hard-coded password has changed, contact the tech support who installed the system or the current admin for the updated login

Inventory System

Admins can:

- View all items
- Track:
 - UPC
 - Product name
 - Weight
 - Quantity

Pantry Analytics Report

Step 1:

Open Report Page

Step 2:

Filter Data (Optional)

- Select:
 - Start Date
 - End Date
- Click Filter Report

Step 3:

View Data

Includes:

- Total inventory items
- Low stock items (≤ 5)
- Visits by major
- Popular items by major
- Total weight distributed
- Full inventory list

Step 4:

Download Report

- Click "Download PDF"
- Includes filtered data

Volunteer Hours Report

Step 1:

Open Volunteer Report Page

Step 2: Filter by Date

- Start date
- End date

Step 3: View Table

Shows:

- Volunteer name
- Email
- Total hours worked

Step 4: Export

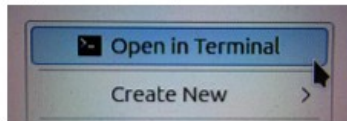
- Click Download PDF

Voiland Food Pantry

Management Software – Operation Instructions (BETA)

STARTING THE WEBSITE:

1. Turn the NUC Computer on.
Username: voilandfp
Password: vfadmin
2. Open the folder PantryApp on the desktop, then 11-FA25-SP26-F-WEB
3. Right click anywhere in the file explorer, and select “Open in Terminal”



4. Type the following command and press enter:
`cd code/backend ; mvn spring-boot:run;`
Wait. If the last line is as follows, it is working:
“Hibernate: select count(*) from admin_users au1_0”
5. WITHOUT closing that window, return to the desktop.
6. Right click anywhere and select “Open in Terminal” once again.
7. Type the following command and press enter:
`firefox --kiosk localhost:8080;`

The website should now be open. Record an error if this does not open the website as intended

USING THE WEBSITE:

If using as a student checking out items, click on "Student Portal"

If using as food pantry staff, click on "Staff/Admin Portal"

CURRENT ADMIN LOGIN (SUBJECT TO CHANGE):

USERNAME: admin

PASSWORD: password123

Right now, functionality exists to manage inventory and view reports on pantry activity. Instructions for admin and student activities exist within the website.

EXITING THE WEBSITE:

1. Press Ctrl, Alt, and F3 at once.
2. If the bottom line is "voilandfoodpantryterminal login: " (IF NOT, GO TO STEP 3), type "voilandfp" and press enter. When "Password: " appears, type "vfadmin" and press enter.
NOTE: any characters typed while inputting the password will NOT appear on the screen for safety reasons, trust that they are still being typed!
3. The bottom line should now be "voilandfp@voilandfoodpantryterminal:~\$". Type "killall firefox" and press enter.
4. Type "logout" and press enter.
5. Press Ctrl, Alt, and F2 at once to exit this screen.
6. Close all existing windows.

If asked if you are sure you want to close a window, say yes.

At this point, the NUC Computer should be ready to be turned off if necessary. The website can be brought back using the "Starting The Website" procedure on the other side of this document.

Instructions provided by Alexander Langland during development. Contact the tech support that installed the system or the current admin if access fails for the updated procedures.

Using the Card Reader

Steps:

1. Ensure the card reader is properly connected
 2. Navigate to the card input section of the application
 3. Swipe the ID card on the reader
 4. Register if a first time user
 5. Save or confirm the entry
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Using the Handheld Barcode Scanner

Steps:

1. Ensure the scanner is powered and connected
 2. Open the item entry or scanning section
 3. Point the scanner at the barcode
 4. Press the scan trigger
 5. Confirm that item data appears in the system
 6. Save or confirm the entry
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Data Entry and Management

- Verify all data before saving
 - Avoid duplicate entries
 - Ensure required fields are completed
 - Log out after use
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Troubleshooting

Login Issues

- Verify username and password

- Check for caps lock
- Contact admin to reset

Card Reader Not Working

- Check connection (USB/Bluetooth)
- Restart the application
- Type or write down information

Barcode Scanner Not Working

- Check power or battery
- Ensure proper connection
- Clean scanner lens
- Verify barcode is not damaged

Data Not Saving

- Confirm database connection
 - Check required fields
 - Refresh browser
 - Contact admin if necessary
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Known Limitations / Notes

(from Kaitlyn Cornish email dated: 4-21-26)

1. Altering UI features (i.e. title)

UI elements such as titles are defined in the front end HTML files. Any changes would need to be made directly in those files, so an admin would need to update the code manually.

2. Editing database data and tables

Database content can be viewed and modified through MySQL Workbench or via the terminal using SQL commands.

3. Logging in and browser usage

The application can be accessed through any web browser set as the default on the user's computer. After running the project locally via the terminal, users can open the provided URL in their preferred browser. Set to Firefox right now.

4. Exporting/printing reports

Reports can be downloaded using the download button on the webpage. The file will save to the computer's Downloads folder, where it can then be opened, printed, or shared, for example, through email.

5. Potential future improvements

One major improvement would be transitioning the application to a cloud-based system. This would allow easier access from any computer without requiring local setup and would improve scalability and usability.

Note: updates are needed for successful online deployment, and updates are recommended for more stable offline deployment. See [Future Testing and Improvements section](#)

Future Testing and Improvements

This section summarizes known areas of fragility, recommended testing procedures, and future improvements identified during project review. It is intended for use by future dev teams and troubleshooters.

1. Scanner Input and JSON Errors

Known Issue

Scanner workflows may fail if malformed or incomplete input is received.

Possible symptoms:

- JSON parsing errors
 - Item submission failures
 - Frontend freeze or invalid state
 - Manual entry failure after scanner interruption
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Recommended Testing

Perform:

- rapid repeated scans,
- disconnect/reconnect testing,
- manual entry after scanner failure,
- invalid barcode testing,
- partial scan testing.

Verify:

- frontend recovery,
 - inventory consistency,
 - proper error messages,
 - successful manual fallback entry.
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Suggested Future Improvements

- stronger input validation,
 - scanner timeout handling,
 - duplicate scan prevention,
 - centralized error recovery.
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2. Multiple User / Inventory Consistency Risks

Known Issue

Inventory updates may become inaccurate when multiple users submit transactions simultaneously.

Possible symptoms:

- incorrect quantities,
- duplicate deductions,
- inconsistent reports.

Recommended Testing

Test with:

- two browser sessions,
- multiple simultaneous checkouts,
- rapid item submissions.

Verify:

- inventory counts remain correct,
 - duplicate submissions do not occur,
 - reports remain accurate.
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Suggested Future Improvements

- transaction locking,
 - concurrency-safe inventory updates,
 - stronger database constraints.
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3. Session and Navigation Stability

Known Issue

Browser navigation and session state may become inconsistent.

Possible symptoms:

- stale pages after logout,
 - broken back-button navigation,
 - scanner focus problems,
 - duplicate submissions.
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Recommended Testing

Test:

- browser back button after logout,
- multiple browser tabs,
- long idle sessions,
- repeated login/logout cycles.

Verify:

- sessions terminate correctly,
 - pages refresh properly,
 - stale data is not accessible.
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Suggested Future Improvements

- stronger session management,
 - cache-control improvements,
 - frontend state cleanup.
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4. Report and Export Reliability

Known Issue

Reports and PDF exports may behave inconsistently with:

- empty datasets,
 - large data volumes,
 - malformed records.
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Recommended Testing

Test reports using:

- empty databases,
- large inventory lists,
- large visit histories,
- date filters.

Verify:

- PDFs generate correctly,
 - exports complete successfully,
 - totals remain accurate.
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Suggested Future Improvements

- query optimization,
 - null handling,
 - export validation,
 - large dataset performance testing.
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5. Security and Deployment Concerns

Known Issue

The current application was designed primarily for local deployment.

Potential concerns include:

- weak default credentials,
 - local-only assumptions,
 - hardcoded configuration values,
 - insufficient production hardening.
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Recommended Testing

Verify:

- default passwords are changed,
 - unauthorized users cannot access admin pages,
 - inventory endpoints require authentication,
 - sessions expire correctly.
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Suggested Future Improvements

Before online deployment:

- remove hardcoded credentials,
 - improve authentication architecture,
 - enable stronger security protections,
 - implement role-based access control.
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Standards Verification and Constraint Testing

Suggested standards for future testing

Recommended Validation Areas

Input Validation

Verify:

- invalid UPC codes are rejected,
 - empty forms are blocked,
 - oversized values are handled safely,
 - malformed requests do not crash the system.
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Database Integrity

Verify:

- duplicate users are prevented,
 - duplicate inventory records are blocked,
 - volunteer hours remain within allowed limits,
 - inventory values never become negative.
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Session and Access Control

Verify:

- users cannot access restricted pages without login,
 - sessions terminate after logout,
 - browser cache does not expose previous user data.
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Hardware Reliability

Verify:

- scanners reconnect correctly,
 - card readers recover after disconnect,
 - manual entry remains functional during hardware failure.
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Performance and Stress Testing

Verify:

- repeated scans do not freeze the system,
 - large reports generate successfully,
 - multiple users do not corrupt inventory.
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Security Best Practices

- Use strong passwords.
 - Do not share accounts.
 - Log out after use.
 - Restrict admin access.
 - Back up the database regularly.
 - Remove default credentials before deployment.
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Support and Contact Information

For assistance, contact:

- Technical Support: kaitlyn.cornish@wsu.edu, brooke.hill@wsu.edu,
alexander.langland@wsu.edu

As of 5 – 7 – 26, this is the dev team. Subject to update.

Revision History

Version	Date	Description	Author
0.1	4-6-26	Initial draft	Jodie Butterworth
0.2	4-23-26	2 nd draft	Jodie Butterworth
0.3	4-27-26	Minor change, 3 rd draft	Jodie Butterworth
.04	5-7-26	4 th draft, final for semester	Jodie Butterworth

Generative AI, ChatGPT was used in making this user guide.